EE1007
Introduction to Computer Science Laboratory

Lab 1
Your First Program

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Lab 1.1

- Install Code::Blocks 20.03 and execute your first program
- What is Code::Blocks?
  - Code::Blocks is a free, open-source cross-platform IDE that supports multiple compilers including GCC, Clang and Visual C++
  - It is developed in C++ using wxWidgets as the GUI toolkit
  - Code::Blocks is being developed for Windows and Linux (also macOS) and has been ported to FreeBSD, OpenBSD, and Solaris

Lab 1.1

- What is IDE?
  - Integrated development environment
  - A software application that provides comprehensive facilities to computer programmers for software development.
  - An IDE normally consists of at least a source code editor, build automation tools, and a debugger
  - Some IDEs contain the necessary compiler, interpreter, or both

Lab 1.1 Steps

- Download and Install Code::Blocks
- Select Compiler
- Create Your First C++ Project
- Build and Run

Download Code::Blocks

- https://www.codeblocks.org/downloads/binaries/

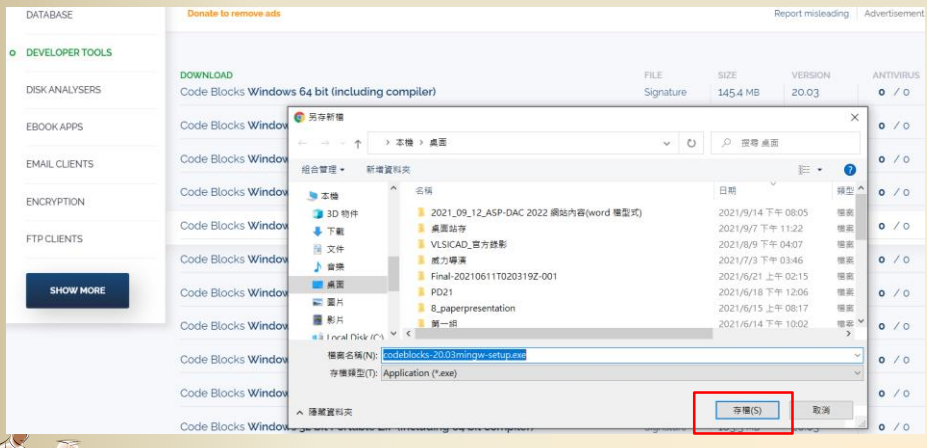
Microsoft Windows

File	Download from
codeblocks-20.03-setup.exe	FossHUB or Sourceforge.net
codeblocks-20.03-setup-nonadmin.exe	FossHUB or Sourceforge.net
codeblocks-20.03-nosetup.zip	FossHUB or Sourceforge.net
codeblocks-20.03mingw-setup.exe	FossHUB or Sourceforge.net
codeblocks-20.03mingw-nosetup.zip	FossHUB or Sourceforge.net
codeblocks-20.03-32bit-setup.exe	FossHUB or Sourceforge.net
codeblocks-20.03-32bit-setup-nonadmin.exe	FossHUB or Sourceforge.net
codeblocks-20.03-32bit-nosetup.zip	FossHUB or Sourceforge.net
codeblocks-20.03mingw-32bit-setup.exe	FossHUB or Sourceforge.net
codeblocks-20.03mingw-32bit-nosetup.zip	FossHUB or Sourceforge.net

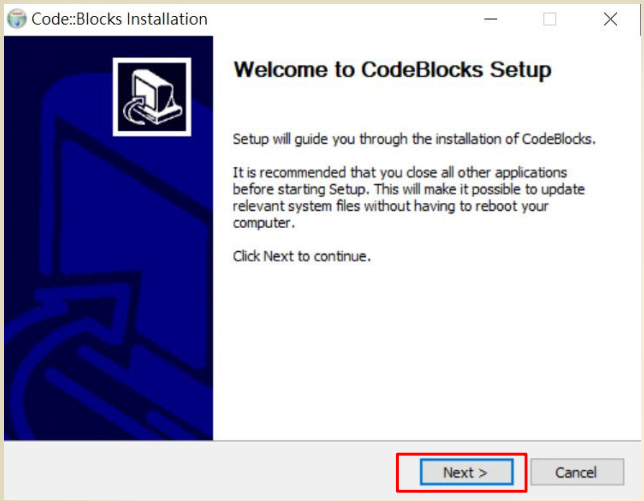
NOTE: The codeblocks-20.03-setup.exe file includes Code::Blocks with all plugins. The codeblocks-20.03-setup-nonadmin.exe file is provided for convenience to users that do not have administrator rights on their machine(s).

NOTE: The codeblocks-20.03mingw-setup.exe file includes additionally the GCC/G++/GFortran compiler and GDB debugger from MinGW-W64 project (version 8.1.0, 32/64 bit, SEH).

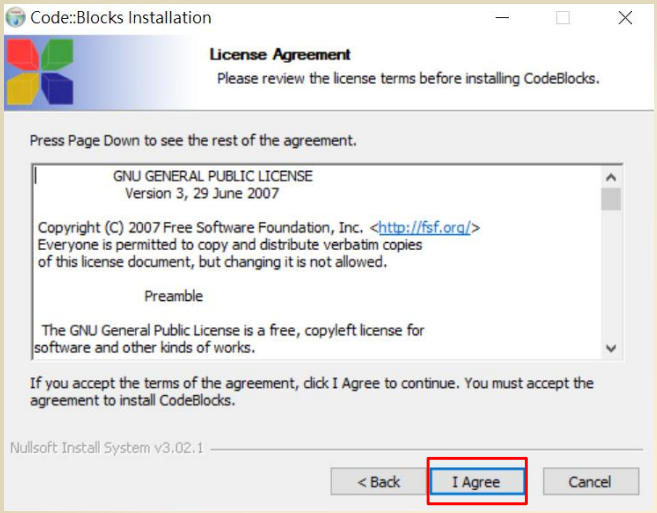
Download Code::Blocks



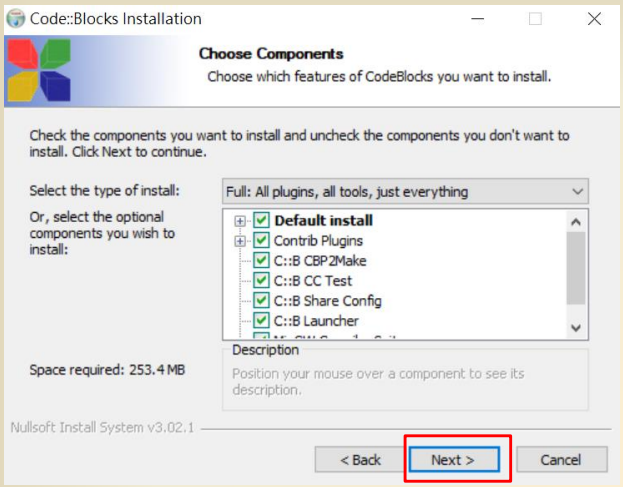
Install CodeBlocks



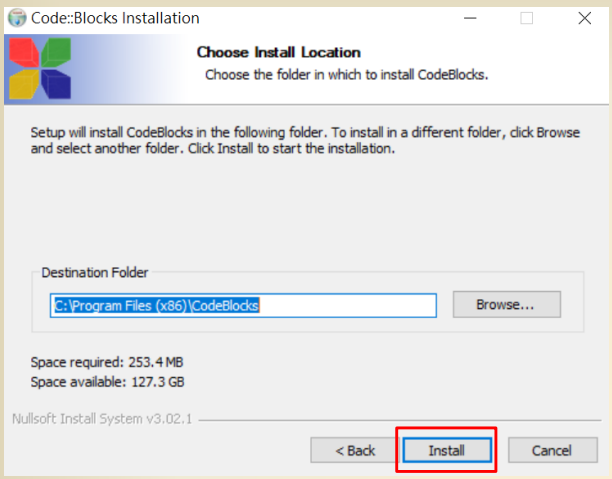
Install CodeBlocks(Cont.)



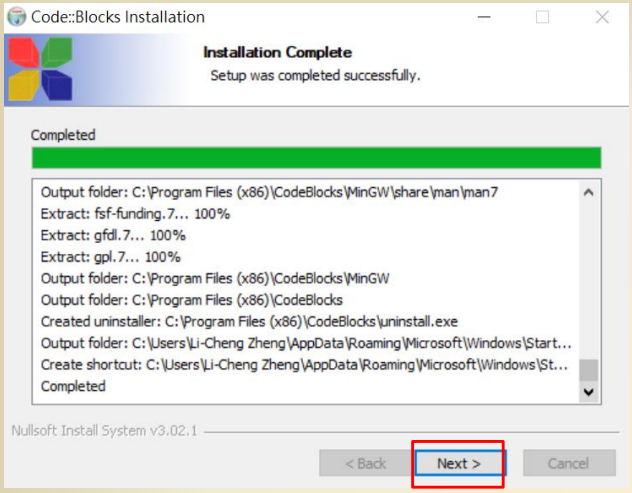
Install CodeBlocks(Cont.)



Install CodeBlocks(Cont.)



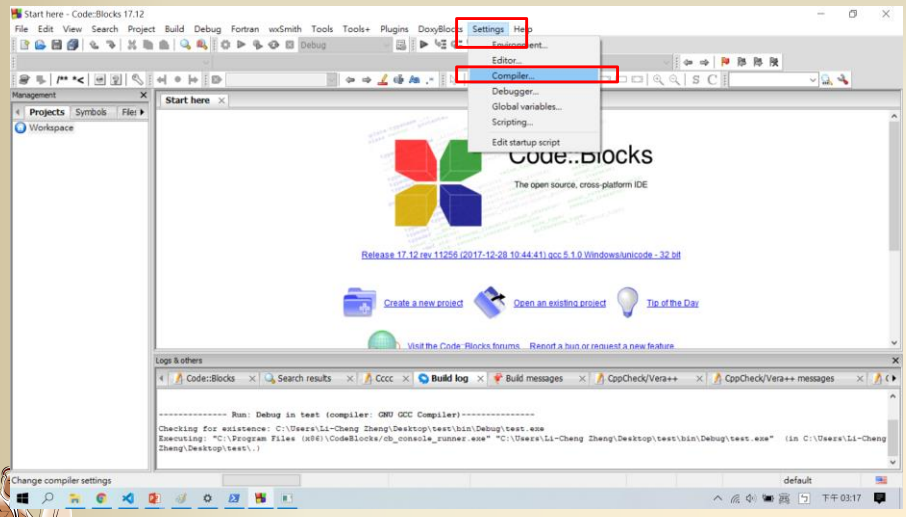
Install CodeBlocks(Cont.)



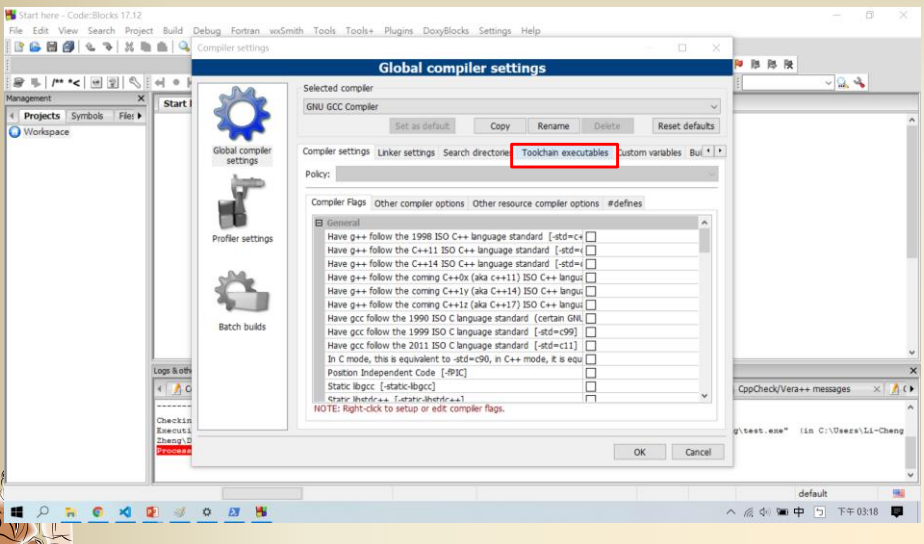
Lab 1.1 Steps

- Download and Install CodeBlocks
- Select Compiler
- Create Your First C++ Project
- Build and Run

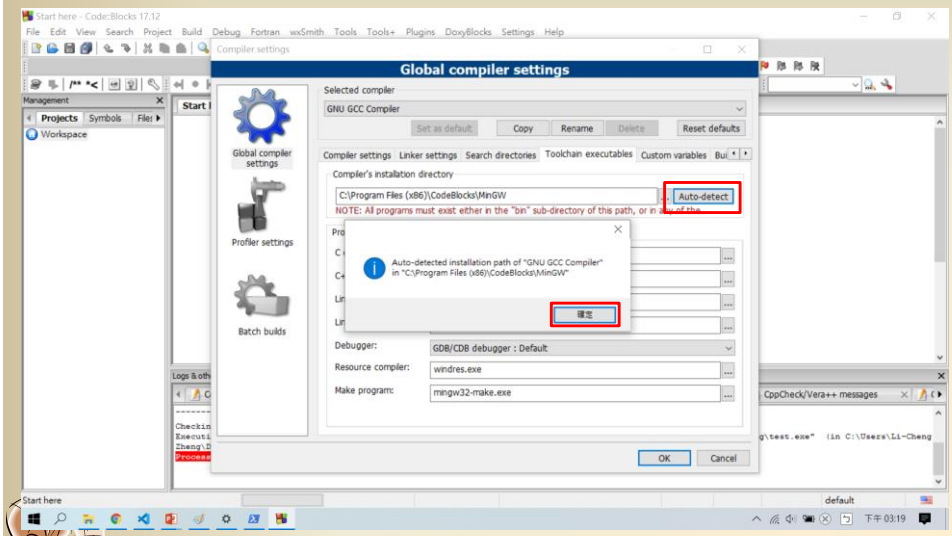
Select Compiler



Select Compiler(Cont.)



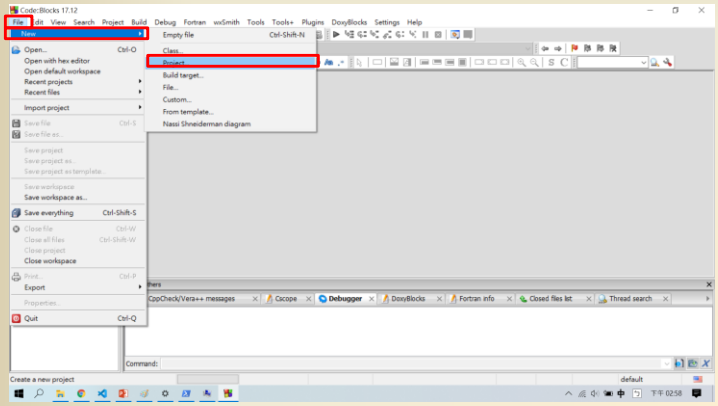
Select Compiler(Cont.)



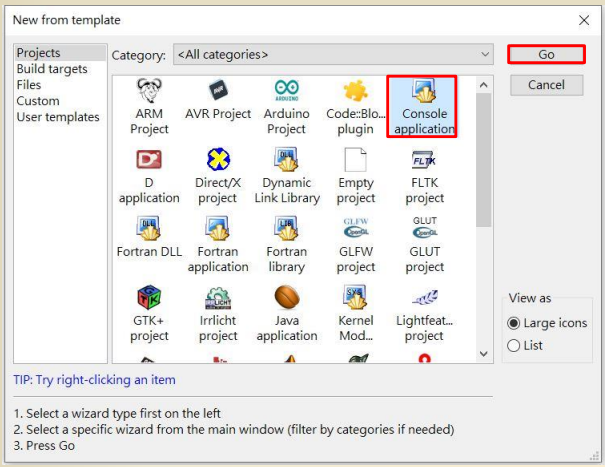
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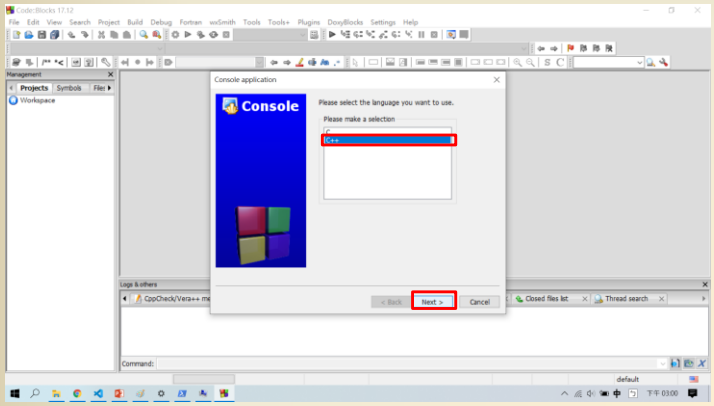
Create Your First C++ Project



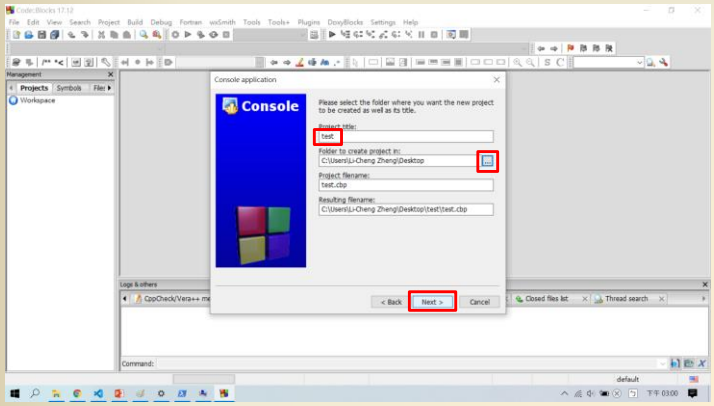
Create Your First C++ Project(Cont.)



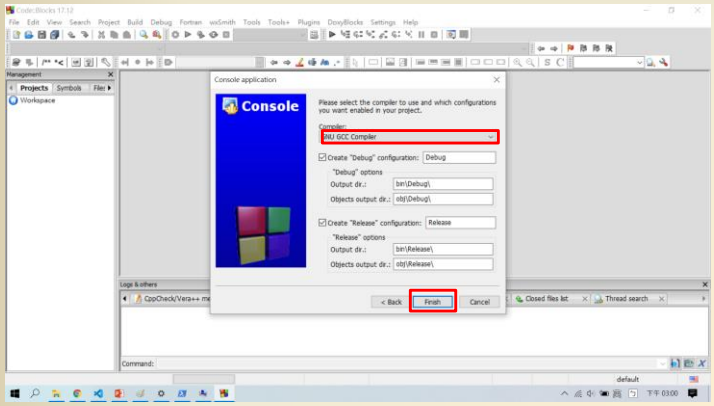
Create Your First C++ Project(Cont.)



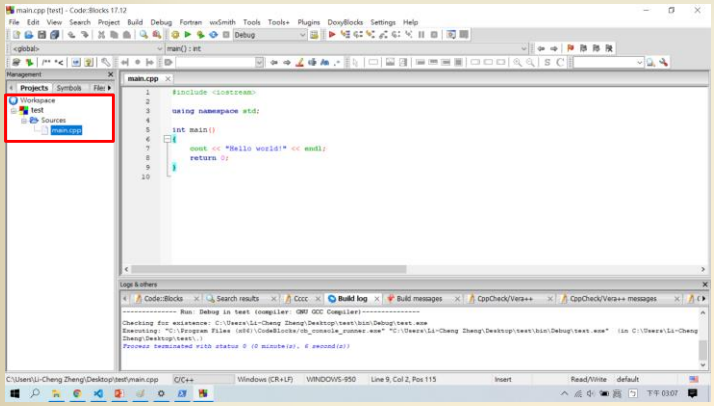
Create Your First C++ Project(Cont.)



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Create Your First C++ Project(Cont.)

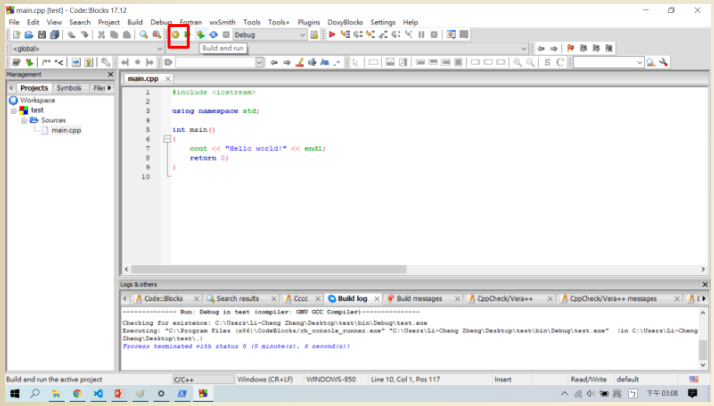


Outlines

- Download and Install CodeBlocks
- Select Compiler
- Create Your First C++ Project
- Build and Run

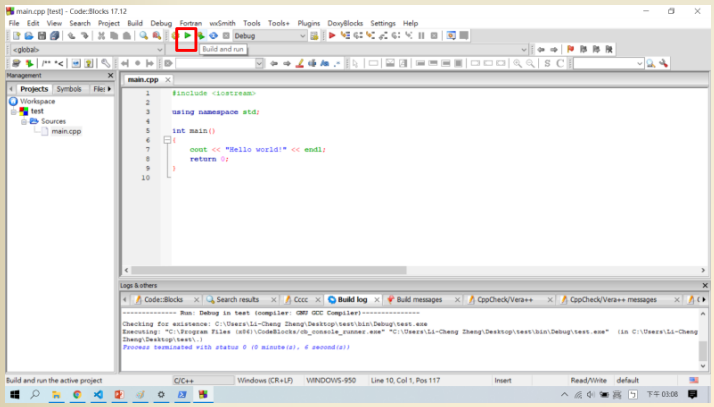
Build and Run

- Build



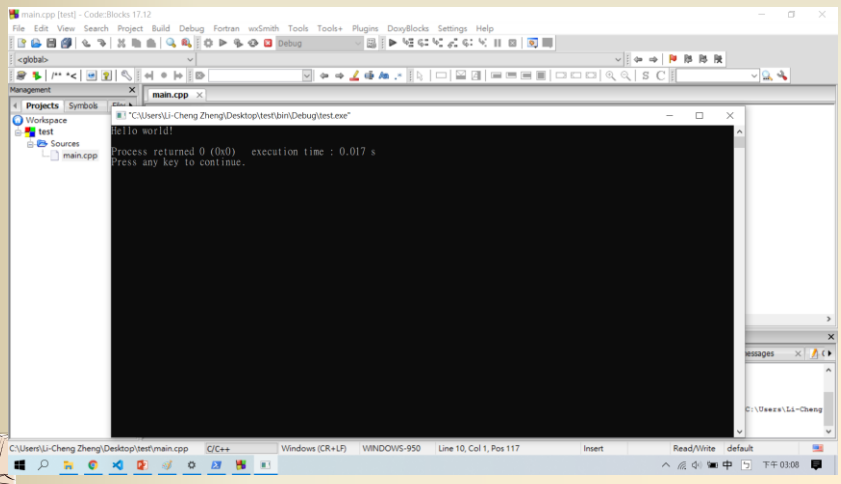
Build and Run

◆Run



Build and Run

◆Execution



Lab 1.1

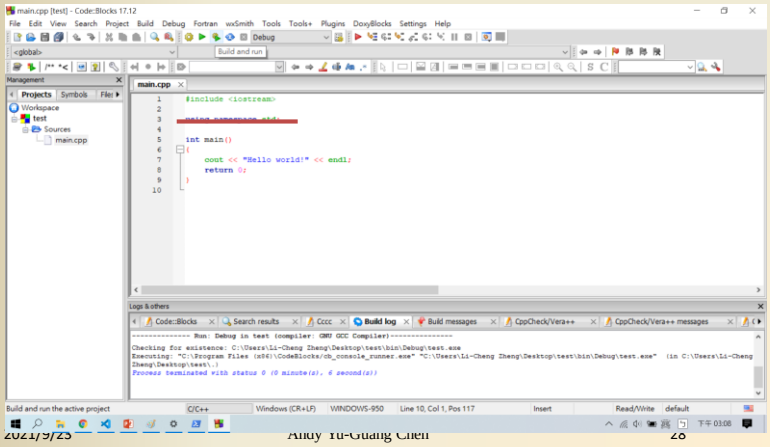
- Then modify the given program to show the following output result

```
C:\Coding_Example\First_prog\bin\Debug\First_prog.exe
Hello
world!

Process returned 0 (0x0)   execution time : 0.055 s
Press any key to continue.
```

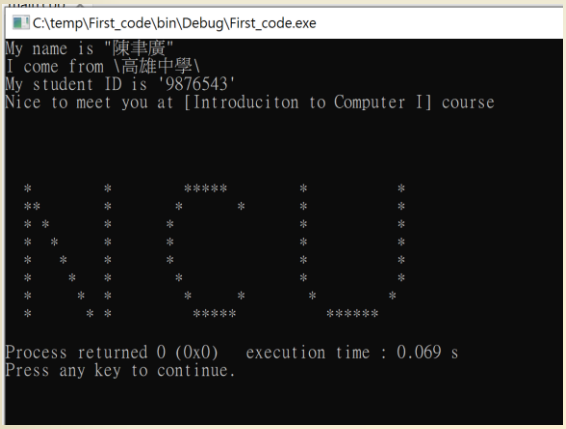

Lab 1.2

- Write your first program to introduce yourself
  - Please remove the line “using namespace std;”



Lab 1.2

- Write your first program to introduce yourself
  - Please print the following words on the screen



Evaluation

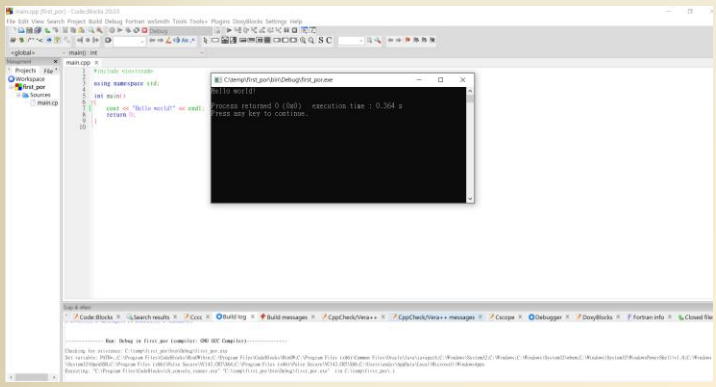
- Lab1.1(40%)
  - The new line character is required between “Hello” and “world!”
- Lab1.2(60%)
  - You need to insert your name, high school name, and student ID and between two double quotations, back slashes, and single quotations, respectively. You will obtain 10% for each line.
  - You need to print out “NCU” like the example. TAs won’t count how many starts you use, but make sure TAs can recognize it’s a “NCU”. 20% for each character.

Lab Operation Process

- Do your lab at your own computer (for both Lab 1.1 & Lab1.2, please create two different projects for each lab)
- When you finish, find your TA at Google Meet and have a demo for both lab
- Submit your files to ee-class
- You are good to go!

File Submission

- Lab 1.1
  - You have to submit a screenshot about your Code::Blocks and execution result



File Submission

- Lab 1.1
  - Name you file as 學號_姓名_LabX.X.jpg
  - Upload to ee-calss 作業->Lab 1.1
  - Only submit your screenshot, no source code is needed



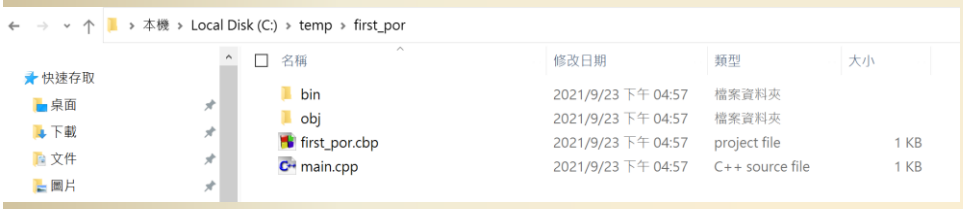
File Submission

- Lab 1.2
  - You have to submit your source code to ee-class
  - Name you file as 學號_姓名_LabX.X.cpp
  - Only source code (.cpp file), not the entire project



How to find your source code?

- Your_project_director >main.cpp
- Copy the file to desktop and rename the file







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